COMMON 3D PRINTING TECHNOLOGIES

3D Printing Technology	Materials	Applications
Sheet lamination methods, such as ultrasonic additive manufacturing and laminated object manufacturing	Paper, plastic, metal, any sheet material that can be rolled	Non-functional prototypes and models, multi-colour prints, casting moulds
Directed metal deposition (DMD) or directed energy deposition (DED)	Metals like cobalt, chrome, titanium	Repairing high-end components, functional prototypes
Material extrusion, most common being fused deposition modelling (FDM)	Plastics and polymers, including polylactic acid (PLA), polycarbonate, acrylonitrile butadiene styrene (ABS), nylon, polyetheretherketone (PEEK), thermoplastic polyurethane (TPU) and thermoplastic elastomer (TPE), carbon fibre filament	Electronic housings, form-fitting fixtures, investment casting patterns
Material jetting, nano-particle jetting, drop-on-demand	Plastics, polymers, silicone, metals	Prototypes, medical models, design presentations
Binder jetting	Polymers, metals, ceramics, glass	Metal parts, low-cost prototypes, sand casting moulds
Vat photo-polymerisation, such as stereolithography (SLA), direct light processing (DLP), and continuous direct light processing (CDLP)	Plastics, polymers, resins	Injection moulds, prototypes and models, dental fixtures, medical parts
Powder bed fusion technologies, including direct metal laser sintering (DMLS), electron beam melting (EBM), selective heat sintering (SHS), selective laser melting (SLM), and selective laser sintering (SLS)	Powder-based polymers and metals, including nylon, PEEK, non-toxic plant-based resin, stainless steel, titanium, aluminium, cobalt, chrome, copper, Alumide (a composite of aluminium and nylon), and Inconel (an alloy of chrome and nickel)	Functional metal parts, low-volume production, hollow designs

to understand diseased conditions better (for drug development) and for potential treatment.

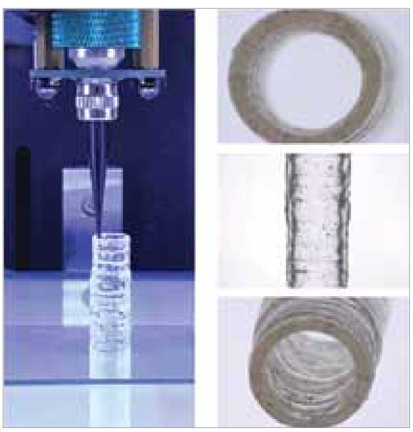
Scientists at Technion-Israel Institute of Technology have created a hierarchical network of 3D-printed blood vessels capable of supplying the necessary amount of blood to an implanted tissue. For a tissue implant to be successful, it must first be permeated by the body's blood vessels for it to receive the required oxygen and nutrients. To achieve this, the tissue is first implanted in a healthy limb of the patient's body, and then shifted to the required location. This complicates and delays the process.

At Technion, the scientists have created a tissue flap in the lab, with all the vessels necessary for blood supply, so it can directly be implanted in the affected area. The new tech has been successfully tested on rats and will be tested on larger animals before it is ready for human use.

While previous experiments used collagen from animals in the bio-ink, here, Israeli company Coll-Plant Biotechnologies has genetically engineered tobacco plants with five human genes to produce collagen! According to the researchers, large blood vessels of the exact shape necessary can be printed and implanted together with the tissue that needs to be implanted. This tissue can be formed using the patient's own cells, eliminating rejection risk.

A team of researchers led by associate professor Dr Akhilesh Gaharwar and assistant professor Dr Abhishek Jain, in the Department of Biomedical Engineering at Texas A&M University, have designed a 3D-bioprinted model of a blood vessel in healthy and diseased states. They have also developed a new nanoengineered bio-ink, which has the strength and optimal viscosity for smooth printing. The ink is printed into 3D cylindrical blood vessels, consisting of living co-cultures of endothelial cells and vascular smooth muscle cells, which maintain integrity of the arterial walls.

The best part is that they have customised a US$200 off-the-shelf thermoplastic printer into a bioprinter, saving thousands of dollars. This scientific development will help in research, treatment, and development of drugs for cardiovascular diseases.



The 3D bio-printed vascular model with native endothelial and vascular smooth muscle cells (Courtesy: Dr Akhilesh Gaharwar)

With calcium phosphate as material, it is now possible to print bones, which unify within weeks after grafting, with low rate of rejection by the body. Quite recently, researchers at Princeton University 3D-printed a bionic ear with above-average hearing abilities, demonstrating the possibility of creating bionic organs by combining biological and nano-electronic functionalities through 3D printing.

Scientists and companies around the world have shown that it is possible to 3D-print several parts of the human body today, from skin to eye lenses and jaw bones. In addition, researchers are also investigating the possibility of bio-ink infused with stem cells, which when injected in a scaffolding could help an organ to grow or regenerate.

Personalised surgical instruments. When compared to exoskeletons and organs, this might see less fantastic, but 3D-printed surgical instruments are equally critical in the life-saving journey. With 3D printing, it is possible to rapidly prototype and develop personalised surgical instruments at a very affordable cost point. Highly miniaturised instruments can also be 3D-printed, helping the surgeon to perform more precise surgeries. Companies like